

Abridged Survey Protocol:

Macrocystis kelp canopy productivity data from BC Central Coast, v1.4.0

Compiled by Ondine Pontier, December 2021

Last updated, Jan 2026

This is an abridged version of the Nearshore Kelp Monitoring protocols available [here](#).

Site Metadata

Monitoring of *M. pyrifera* is conducted at 7 permanent and several additional non-permanent sites near Calvert Island, British Columbia (Map 1, Table 1).



Map 1. Hakai Nearshore *Macrocystis pyrifera* study sites. Coordinates for HIRMD specific sites are available upon request.

Table 1. Site GPS coordinates, frequency of sampling and fixed depths.

Site	Latitude (N)	Longitude (W)	Survey Frequency	Year	Fixed Depth
Triquet	51.8036	-128.2542	3-5 visits/year 1 visit/year	2014-2018 2019-current	4m
Womanley	51.8023	-128.2003	3-5 visits/year 1 visit/year	2017-2018 2019-current	5m
Westbeach	51.6492	-128.1519	3-5 visits/year 1 visit/year	2016-2018 2019-current	2m
Golden	51.9031	-128.2130	3-5 visits/year 1 visit/year	2014-2015 2016-current	4m
Simmonds	51.9619	-128.2619	3-5 visits/year 1 visit/year	2014-2015 2016-current	5m
Stryker	52.0836	-128.3646	3-5 visits/year 1 visit/year	2014-2015 2016-current	4m
Meay	51.6645	-128.0955	3-5 visits/year 1 visit/year	2014-2016 2017-current	2m
Edna	51.8126	-128.2228	1 visit	2019	NA
McMullins_center	52.0492	-128.4038	1 visit	2019	NA
Triquet_beach	51.8045	-128.2540	1 visit	2019	4m
Westbeach_north	51.6562	-128.1460	1 visit	2019	2m
Manley	51.8012	-128.2000	1 visit/year	2019-current	4m
Triqster	51.79817	-128.24019	1 visit	2020-2021	4m
Lyte	51.81435	-128.23700	1 visit	2021	NA
Triquetta	51.79739	-128.23840	1 visit/year	2022-current	4m
Stryker_HIRMD	AUR	AUR	1 visit/year	2021-current	6m
Simmonds_HIRMD	AUR	AUR	1 visit/year	2021-current	4-5m
Spider_HIRMD	AUR	AUR	1 visit/year	2021-current	4m
McMullins_north_HI RMD	AUR	AUR	1 visit/year	2022-2023	5-6m
Goose_HIRMD	AUR	AUR	1 visit/year	2022-current	3-4m
McMullins_HIRMD	AUR	AUR	1 visit/year	2024-current	2m
Cape mark_HIRMD	AUR	AUR	1 visit/year	2024-current	4m
Ivory_HIRMD	AUR	AUR	1 visit	2024	3m

*AUR (available upon request)

Site Layout

Sites were selected on semi-exposed rocky reefs in beds dominated by *M. pyrifera* occurring within a depth range of 2-5m below Mean Lower Low Water (MLLW). Sites have fairly even or gently sloping depths throughout the monitoring area.

Sites were sampled at a range of intervals year round 2014-2018 with higher frequency during the growing season. 2018 onward, sites are visited on an annual basis. Permanent sites have three 20m leadlines placed parallel to one another and approximately 4m apart in the center of the kelp bed (Figure 1). Non-permanent sites have a similar configuration (except for Edna, single transect) but the length of the transects vary between 20, 30 and 60m based on the site.

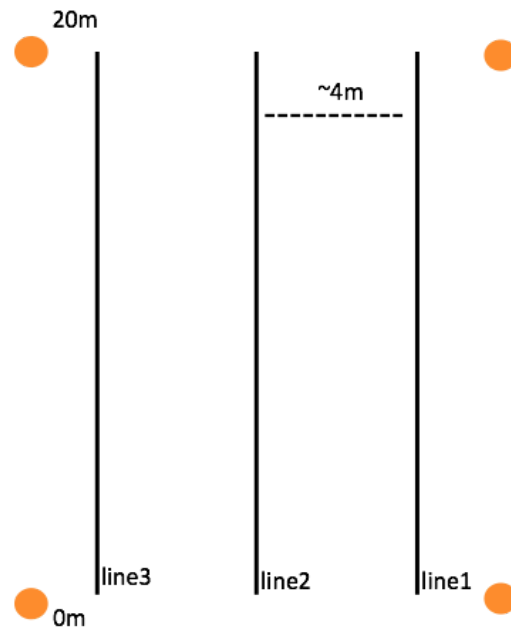


Figure 1. Example of a permanent plot layout at a site (plot size ~ 10m x 20m). Orange buckets mark the four corners of the plots. Plots contain three parallel leadlines (20m) that are 4m apart.

Protocols

The following field protocols are designed to capture measurements necessary for calculating *M. pyrifera* density (plants m⁻²), biomass (kg m⁻²), and productivity (kg m⁻², d⁻¹). **Bold** parameters relate to nomenclature in the dataset file, “**macro_metrics.csv**”. These protocols are based on those developed and used by the Santa Barbara Coastal Long-Term Ecological Research (LTER) program (<http://sbc.lternet.edu/>).

1. Density and size-distribution

Every adult plant (>1m) within 1m on either side of the permanent lines is measured by pulling it down in 1m arm length increments. The plant's holdfast position is first recorded and includes:

- a. The transect number (**Transect**)
- b. The transect length (**Transect_length**), which is how much of the transect tape or permanent line will be surveyed and may vary
- c. The side of the transect (**L/R**), established by facing the end (20m mark) of the transect line
- d. The distance along the transect (m) (**Distance**)
- e. The depth (**Depth**) of the holdfast as read by the divers computer depth gage
- f. The fixed depth (**Fixed_depth**) of the site, established prior to surveying and based on Mean mean water levels.
- g. Number of fronds 1m (**Fronds_1m**) above the holdfast (0 if kelp does not reach 1m)

If the plant does not reach the surface, **Subsurface canopy (Fig. 2)**, record the following:

- a. The mean length (**Mean_F**) of all fronds on the plant
- b. Record a “0” for fronds at the surface (**Fronds_SFC**) and a “0” for **SFC_length**

If the plant reaches the surface, **Surface reaching kelps (Figure 2)**, record the following:

- a. Number of fronds 1m above the holdfast (**Fronds_1m**).
- b. The number of fronds that reach the surface (**Fronds_SFC**). This is done based on a predetermined site-specific fixed depth 2016 - current (**Fixed_depth**). Prior to 2016, this was done using the dive computer depth reading at time of survey (**Depth**).
- c. The surface length of longest surface canopy frond (**SFC_length**).
- d. Because holdfasts of *M. pyrifera* can be large, sometimes part of a holdfast will be within the 1m sampling swath. In this case, if the plant is only partially in the 1m swath, put a “P” in the **C/P/T** column, and only count the number of fronds at 1m, nothing else. If the plant ends abruptly at the surface, put a “C” in the **C/P/T** column, and record all measurements as usual. If the plant is tangled, record a “T” in the **C/P/T** column and record **Fronds_1m** as well as any other measurements the diver is able to obtain (to the best of their ability).

In 2021 and onward, in situ measurements for estimating individual plant biomass were reduced to the number of fronds 1m above the holdfast (**Fronds_1m**) for all sites. This data subset is found in the dataset file, “**macro_density.csv**”.

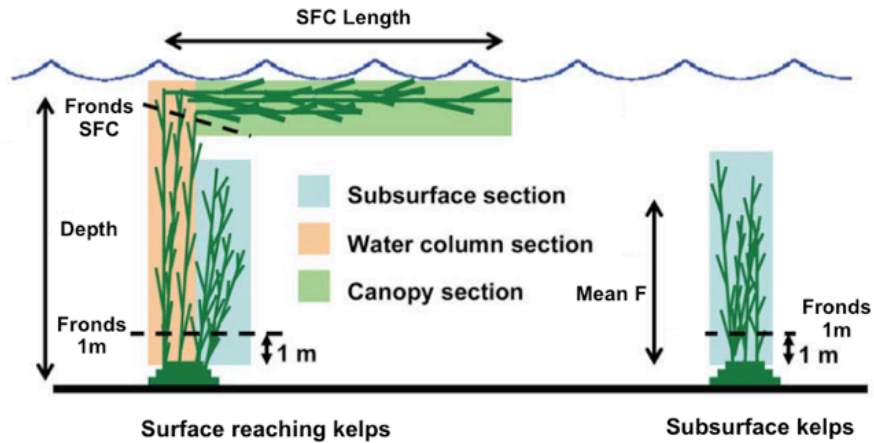


Figure 2. Diagram showing the two different categories of *M. pyrifera* kelps. Surface reaching kelps have floating surface canopies, while the fronds of subsurface kelps don't reach the surface.

2. Plant and frond loss (only seasonal surveys)

Twelve randomly selected plants are **Tagged** using zipties and unique numbered tags. In addition to the measures described above, the following is recorded for each of these twelve plants (also included in “macro_size.csv”):

- Tag #**
- Total number of sporophylls (**Sporo**)
- Number of reproductive sporophylls (**Sporo_R**)
- Average % of blades covered by sori of reproductive blades (**Sporo_P**).

In addition, a subset of six plants are selected (2 per line) for which every individual frond is tagged using a ziptie and the following is recorded:

- Number of tagged fronds present on arrival (**Fronds_exist**)
- Number of newly tagged fronds (**Fronds_new**)

3. Harvest

Five plants are collected at random outside of the monitoring plot once a year (July or August). Each of the plants are separated and every frond is measured in meters and weighed in kilograms on the dock or boat. Pieces of stipe and blade are put aside for wet to dry regressions.

Note: Starting in 2019 fronds were separated into two sections: subsurface or surface (**SUB** or **SFC**) to account for the difference in material biomass below and above the surface.

4. Wet Dry

A piece of stipe and blade is kept and dried during harvest to obtain a wet to dry biomass conversion factor in the following manner:

- Empty trays are numbered with **sample IDs** and pre-weighed on analytical balance (**tray**). A **Sample ID** is allocated chronologically based on pre-existing data.

- b. An additional **unique ID** is allocated to the sample if it will be kept for isotope analysis. These are Nearshore wide IDs and are kept track of in the Nearshore Isotope dataset.
- c. The **tag** number for that plant is also recorded if known.
- d. The sample is cut (stipe 5cm length, blade 5 x 5 cm square) and recorded as well as the type of material (**stipe_blade**).
- e. The trays containing wet materials are weighed (**tray_wet**).
- f. The samples are dried in the drying oven overnight (12 hour minimum). Additional time may be required if the sample is not dry.
- g. The trays containing the dried material are weighed (**tray_dry**).

Amendments and Additional Biomass Surveys

In summer 2018 and 2019, additional density surveys were conducted and a subset of the protocols were used in between transects at pre-existing permanent sites as well as additional sites. In 2021 and onward, in situ measurements for estimating individual plant biomass were reduced to the number of fronds 1m above the holdfast (**Fronds_1m**). The full suite of measurements for plant biomass estimates were used at each of the HIRMD sites during the 2024 summer surveys.

1. 2018 Permanent Sites Protocols

Record the position of any plant between line 1 and 2 (e.g., Fig.3, Line 1.5) that did not fall within 1m on either side of the lines. Count number of fronds present at 1m above the holdfast (**Fronds_1m**). Repeat this between line2 and line3 (e.g., line 2.5).

If frond density is too high to keep track of, separate area into 2m long bins and count the number of fronds within each bin.

If leadlines are too far apart and one cannot keep track of plants easily, subdivide the middle section, example Figure 4.

If leadlines do not align, base bins off of center line (Line 2), example Figure 4.

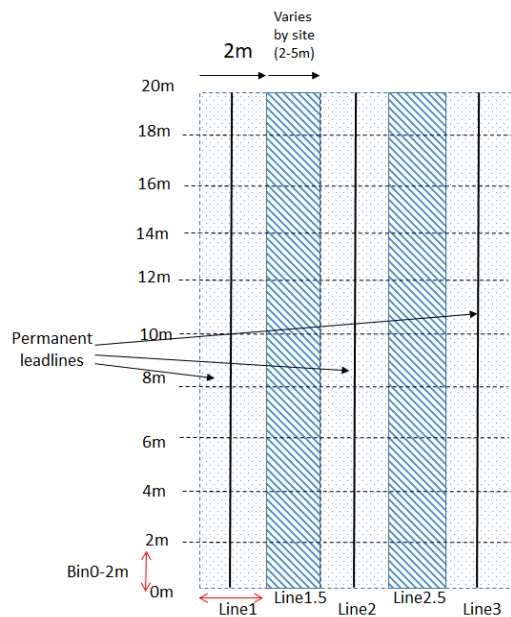


Figure 3. General example of permanent site layouts for density counts.

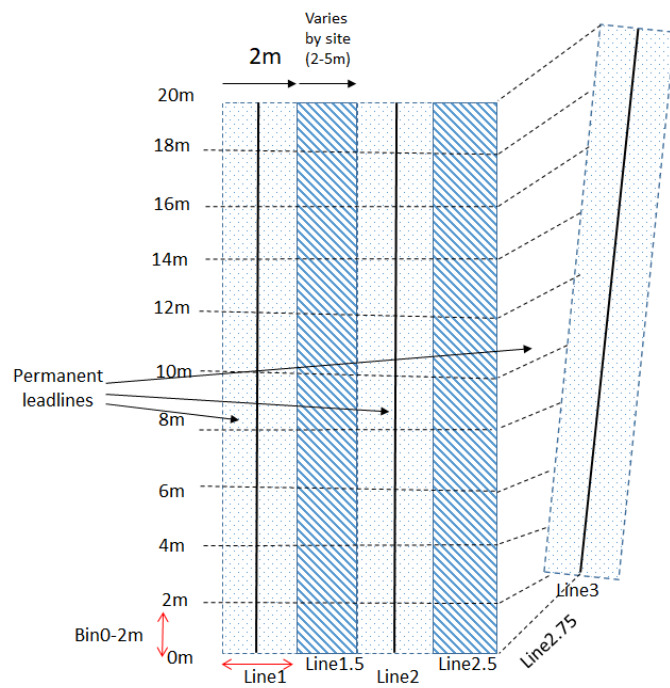


Figure 4. Example of permanent site layout where lines do not align for between-line frond density counts. Line 2.75 (white area) is added to account for all plants between Lines 2 and 3.

2. 2019 Additional Site Protocols

Four non-permanent sites were surveyed using 2m swaths (n = 1-3 per site) of various lengths and configurations (Table 1). *M. pyrifera* density was recorded every 2m and a random sample of plants were harvested along the swaths.